

# STAPP: Designing a Tool for People with Korsakoff's Syndrome to Re-learn Daily Activities Step by Step

Sanne Beijer

Department of Industrial Design, Eindhoven University of  
Technology, Eindhoven, Netherlands  
s.g.beijer@tue.nl

Maarten Houben

Department of Industrial Design, Eindhoven University of  
Technology, Eindhoven, Netherlands  
m.houben1@tue.nl

Fenna Dam

Luckt B.V., Eindhoven, Netherlands  
f.dam@luckt.nl

Rens Brankaert

Department of Industrial Design, Eindhoven University of  
Technology, School of Allied Professions, Fontys  
University of Applied Sciences, Eindhoven, Netherlands  
r.g.a.brankaert@tue.nl

## ABSTRACT

Korsakoff Syndrome (Korsakoff) is a brain disorder that causes severe brain impairments that have consequences for executing activities of daily living (ADL) and often requires lifelong assistance in these activities. Assistive technology can be beneficial for supporting both formal caregivers and people with cognitive impairments. However, as there is little research into how assistive technology can be applied in the context of Korsakoff, more insight is needed into how assistive technology can support the specific needs of people with Korsakoff in re-learning ADL. This paper presents a case study in which a mobile application (i.e., STAPP) is developed through an iterative design process with people with Korsakoff, formal caregivers, designers, researchers, and developers. The application focused on two ADLs (e.g., coffee making and laundry). We report on the design process, the application, autonomy, lack of in-person support, and the importance of in-context instructions.

## CCS CONCEPTS

• **Human-centered computing**; • **Human-computer interaction (HCI)**; • **Empirical studies in HCI**;

## KEYWORDS

Participatory Design Research, Korsakoff's Syndrome, Assistive Technology, Activities of Daily Living

## ACM Reference Format:

Sanne Beijer, Fenna Dam, Maarten Houben, and Rens Brankaert. 2024. STAPP: Designing a Tool for People with Korsakoff's Syndrome to Re-learn Daily Activities Step by Step. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3613905.3637119>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).  
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA  
© 2024 Copyright held by the owner/author(s).  
ACM ISBN 979-8-4007-0331-7/24/05  
<https://doi.org/10.1145/3613905.3637119>

## 1 INTRODUCTION

Care organizations face many challenges today such as the aging population, financial pressures, shortage of staff, and higher workloads in the care sector [8, 12]. Moreover, care staff are providing increasingly complex care due to the complexity of residents' health and social needs [8, 20]. As a result, the provision of care for people with cognitive challenges such as dementia and Korsakoff's Syndrome (Korsakoff) is affected.

Korsakoff is a brain disorder predominantly caused by a thiamine (vitamin B1) deficiency [15]. This thiamine deficiency is often a result of alcohol abuse and subsequent malnourishment [13], but there are also other known cases of non-alcoholic thiamine deficiency often caused by malnutrition [10]. Korsakoff can severely impair the brain and brain functioning, resulting in all kinds of cognitive and behavioral challenges. For example, several memory-related challenges, apathy (e.g., lack of initiating behavior or activities), confabulations (i.e., 'lying' to cover up gaps in knowledge, or making sense of unfamiliar situations), executive dysfunction (e.g., problems with planning), and emotion perception [1]. These challenges have far-reaching, permanent, effects on the activities of daily living (ADL) of people with Korsakoff, and lifelong assistance is often necessary [1, 3].

Challenges with ADL are often reinforced by a lack of awareness of having Korsakoff Syndrome (i.e., limited insight into their illness) [1]. This limited insight results in the belief of people with Korsakoff that they can execute ADL, even though in reality they need assistance, which makes learning difficult [1]. Even though learning is difficult, there is evidence that cognitive rehabilitation and learning are possible for people with Korsakoff [3, 14–16, 21]. Focusing on re-learning activities of daily living may increase the patient's functional autonomy and might reduce care needs. Moreover, interactive technologies provide novel opportunities to support people with Korsakoff and their caregivers. In addition, an increase in autonomy for people with Korsakoff may reduce the pressure on care provider services that are currently resource-intensive, and already under pressure [11].

In this paper, we present a case study about STAPP: an interactive digital application for a smartphone to aid people with Korsakoff to execute daily tasks independently. We developed this application through an iterative design process involving people with Korsakoff, formal caregivers, designers, researchers, and developers [4]. STAPP



**Figure 1:** –STAPP is a digital smartphone application (a) that provides step-by-step instructions with a voice-over for people with Korsakoff (b) who can place the smartphone in its holder (a) near the household appliance they want to use (c).

(figure 1), supports a person to complete multistep activities that contribute to someone’s independence (i.e., making coffee, and doing laundry) and thereby increase their autonomy in everyday life.

In this case study, we explored how technology supports formal caregivers and people with Korsakoff, during their daily activities. We report on the design process and the preliminary findings of the evaluation of STAPP. We found that 1) STAPP has the potential to increase autonomy, 2) formal caregivers missed in-person support, and 3) that in-context instructions are important. Based on these findings, we formulated lessons learned. Overall, this case study illustrates the development and experience of a digital application that taps into the ability of people with Korsakoff. On top of this, we focus on the qualitative experience of assistive technology and expand the literature about assistive technology in the context of Korsakoff Syndrome, which is still in its infancy.

## 2 RELATED WORK

In literature, there are two common approaches in applying learning techniques in the rehabilitation of people with Korsakoff: 1) the deployment of external memory aids (e.g., prompt cards, leaflets) and supportive tools (e.g., electronic diaries, personal digital agents), and 2) the use of ‘*errorless learning*’ [3, 14, 15]. *Errorless learning* is a learning method to prevent mistakes during the learning process by providing feedforward instructions before action and splitting up tasks into small actionable steps [16]. The errorless learning method prevents the learner from making mistakes during the execution of the task. At each step, the learner receives adequate cues to continue [15].

In HCI, there are examples of applications that focus on stepwise learning in dementia research. For example, a VR environment provides preliminary support in the training in ADL such as making coffee to improve the performance of everyday actions of people with dementia [21]. In addition, breaking down a complex task into

smaller steps with easy-to-understand visuals and a well-organized layout contributes to the autonomy and self-efficacy of people with dementia [17]. Even though the insights from dementia research serve as inspiration to the implementation of stepwise learning, these insights are not directly translatable to people with Korsakoff as they have a more severely impaired short-term memory and are often more digitally literate than the general dementia population [1, 13]. Biemond and colleagues [3] pioneered by showing that people with Korsakoff can successfully re-learn activities of daily living through an errorless learning application. They mention that the errorless learning application 1) possibly reduces time and resources for successful implementation and 2) the benefit of having a fixed structure for each training session in clinical practice. Even though the knowledge of how to apply learning techniques has increased, relatively little is known about the translation of these techniques into assistive technologies.

## 3 THE DESIGN PROCESS OF STAPP

The project was a collaborative effort with Pleyade, a Dutch care organization, conducted at their specialized care ward for people diagnosed with Korsakoff. The residents with Korsakoff receive daily care and are still allowed to leave the ward independently if deemed safe by the care staff. The project was initiated by the innovation team of the care organization in which the Eindhoven University of Technology partakes. Under the guidance of a professional designer (second author) and researcher (first author), an iterative and participatory process [4] has been conducted for approximately a year. In multiple iterations, people with Korsakoff, formal caregivers, designers, researchers, and developers were involved through co-creation workshops and interviews. The design process of STAPP contained three phases: 1) empathizing, 2) ideating, 3) developing and evaluating. During the third phase, we improved the developed concepts based on evaluation, including two informal feedback

sessions with residents to iterate and improve the application, and a third evaluation over three weeks.

### 3.1 Empathizing

During the first phase, the first author gathered information about the care ward, its employees, and its residents. The employees reported that residents have limited to no insight into their disease; they think they live temporarily in the ward while the ward is their new home. The concept of 'home', and equal partnership was questioned by the formal caregivers. They raised the question: *"Are people really home when there is someone who checks on you and reminds you of doing tasks?"*. In addition, we found that the residents have trouble with initiating daily activities. An important question raised by formal caregivers was *"How can we guide people in their daily activities without taking over control of resident's lives?"*. Especially because there is little time and few resources available to support residents with Korsakoff more actively [22]. In this specific ward, there are two formal caregivers for twelve residents, resulting in limited resources for personal guidance. Under the guidance of the innovation team, the design question was formulated: How can we help residents with daily planning, taking initiative in activities of daily living, and remembering appointments in an equal manner?

### 3.2 Ideating

During the second phase, co-creation sessions [4] took place under the guidance of the innovation team in the care facility with eight formal caregivers. During these sessions, formal caregivers brainstormed on design solutions regarding guidance, planning, and initiating activities. One design solution that was proposed by the formal caregivers was an application that supports planning, daily activities, and learning. After the co-creation sessions, the second author created concept sketches of the design solutions, showed these sketches to the residents, and asked about their opinions. Based on the results of these co-creation sessions and the residents' opinions, the design question was rephrased by the care innovation team: *"How can we help residents perform tasks more autonomously, in an equal manner?"*, and STAPP was developed into a working digital application.

### 3.3 Developing and Evaluating

In the third phase, the innovation team developed STAPP with the support of the care professionals from the ward. STAPP aims to support people living with Korsakoff to execute and learn the steps to use everyday household appliances, thereby contributing to their autonomy. Within this project, we focused on two activities that were proposed by formal caregivers: 1) coffee making and 2) doing the laundry. The two activities were split into small, actionable, steps [16]. To start the application for a specific device, the mobile phone had to be placed in a wooden holder with an NFC tag specific to that device (Figure 1). Making coffee was split up into 18 steps starting with *'place the phone in the holder'* and closing with *'put the filters back into the cabinet'*. Similarly, we created 21 individual steps for using the washing machine, as illustrated in Figure 2. The steps of making coffee and doing laundry were translated into short videos with corresponding audio assignments to instruct the user before an action. The digital application provides an interface

to navigate through these instruction videos in a simple way to provide some control for the user (Figure 3).

During two informal feedback sessions, three residents tried the application while the second author and a formal caregiver observed the interactions. From these iterations, we learned that unclear instructions would induce stress. For example, in one case, the voice-over said: *"pour the water into the reservoir"*. The participant poured the water into the wrong reservoir, making him unable to make coffee and causing confusion. When the voice-over was later changed into *"open the lid and pour the water in the back of the coffee machine"*, the resident was able to perform this step successfully on their own without stress.

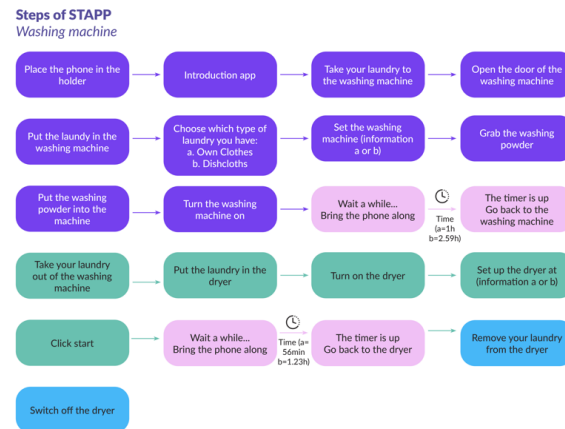
We chose to make an application because it is easily accessible for both formal caregivers and people with Korsakoff. Research has also demonstrated increased mobile phone usage by people with mild cognitive impairment [7] and illustrated that people with Korsakoff can use smartphones as electronic aids [13]. The application was made available for phones running on Operating System Android. To use the application, the phone needed to be placed in the physical holder containing an NFC tag (Figures 1 and 3). The holder would be fixed in place near the designated appliance. When the user has launched the digital application and the phone is put into the holder, the phone automatically reads the NFC tag that is embedded into the holder and starts STAPP for the specific appliance it is next to.

## 4 EVALUATION METHOD

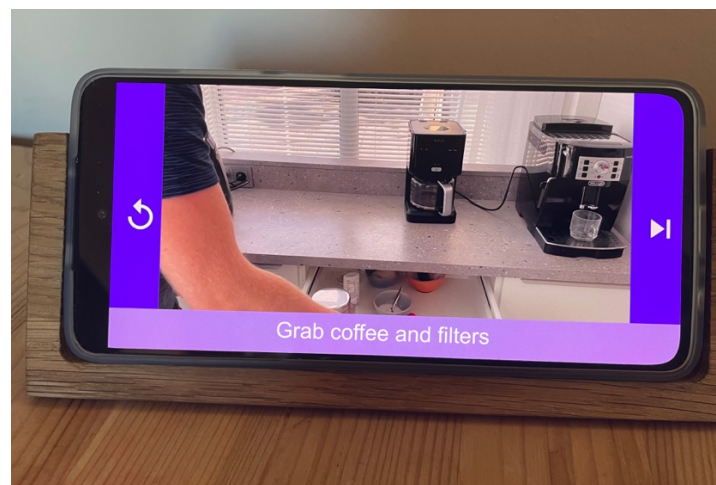
In the next sections, we use pseudonyms for the participant's names. We evaluated the experience of STAPP over three weeks. STAPP was used by Theo, a resident living with Korsakoff. Theo is 57 years old and was diagnosed with Korsakoff's disease due to complications resulting from a stomach reduction and insufficient intake of vitamin supplements. The nature of Korsakoff makes it challenging for residents to recall and reflect on the use of STAPP over time, due to confabulation and the inability to reflect on the use due to severe short-term memory. Therefore, we relied on observations, field notes, and reflections to capture the care perspective on the benefits of STAPP for people with Korsakoff.

For the evaluation, a five-minute interview was conducted every week over the phone with Paul, the formal caregiver of Theo. In this interview, observations and experiences regarding STAPP were discussed. After three weeks, in-person semi-structured interviews were conducted with Paul, Cindy, and Lucy, the formal caregivers of Theo. In this exit interview predefined topics were discussed, such as 1) The formal caregiver's and Theo's attitude towards the application, 2) how this changed through usage, 3) when or how the application was used, 4) what was easy and difficult in using the application, and 5) what should be changed. Besides the interviews, an observation scheme was filled in by two other formal caregivers (C1 & C2), that asked about 1) the date, 2) which activity (i.e., coffee or laundry), 3) what went well/wrong, and 4) feedback for the designer.

The interviews of the three formal caregivers (Paul, Cindy, and Lucy), were recorded, transcribed, coded, and analyzed by the first two authors through reflexive thematic analysis [5]. The transcriptions of the interviews and pictures of the filled-in observation



**Figure 2:** – STAPP provides step-by-step instructions for activities of daily living, for example, using a washing machine. STAPP describes the steps of putting laundry in the washing machine, waiting for the laundry to be finished, drying clothes, waiting for the dryer, etc.



**Figure 3:** – The interface of STAPP provides a button on the left to replay the video, a button on the right is for navigating through the different steps while the text in the lower bar shows the step. The video plays automatically with a voice-over that gives specific instructions, and the video can be paused by touching the middle screen.

scheme were printed, cut into smaller chunks of data, and categorized per subject. Subsequently, all co-authors were involved in iterating and discussing the sub-themes leading to the final themes that will be presented below.

This research was conducted as part of an innovation trajectory in this specific care ward that was initiated by the organization's innovation team. The innovation team is experienced with similar innovation projects and conducts practice-based research in the care context of the organization. Informed consent was obtained from Theo and the formal caregivers by the second author. In addition, the consent form, the goal and duration of the study, and the data processing were discussed by the designer, researcher, Theo,

and the formal caregivers before participation. The inclusion of residents in this study was assessed by formal caregivers who provide daily care to the residents. The professional caregivers were always present during interaction with the application. Because people with Korsakoff are a sensitive target group, and may not remember giving consent in the past, or shift quickly in their emotions, the informed consent was a continuous dialogue between residents, caregivers, and researchers [6].

## 5 FINDINGS OF THE EVALUATION

### 5.1 Increasing Autonomy

The formal caregivers observed how the application could increase the independence of residents and may help with initiating activities. As Paul reported: *"Someone can do things individually, instead of having someone watching over their shoulder and correcting them. I do not think it is nice when someone is standing behind you, telling you what to do. Doing it by yourself, without help. . . eventually, someone can make coffee without help"*. This was agreed upon by Lucy: *"(. . .) When he can make coffee himself, I don't have to do that anymore, so it is a win-win situation"*. The application can also provide a moment of contact and social activity: *"Yes, it is nice to do something together, to have contact with someone"* (Cindy). However, both Lucy and Paul mentioned that the application is purely beneficial for Theo in increasing autonomy and self-confidence, as it does not save time in caregiving tasks directly.

The caregiver's observations revealed that Theo was able to operate the smartphone and follow the instructions easily. Paul observed that in the beginning, the navigation in between videos was not always clear for Theo, even though they seemed straightforward for Paul. Only after listening to the voice-over that told him to push the arrow, Theo was able to continue. *"I noticed that when I asked him to, first I had to explain what the smartphone was for, but now I only show him the smartphone and he knows what to do. This enabled me to loosen my grip. I leave him with the phone to let him figure out the steps. And he succeeded every time"* (Paul). After incorporating the application into the routine, no questions were asked, and Theo immediately understood what was expected of him.

Even though the interaction was straightforward, Paul faced difficulties while introducing the application. As explained before, one symptom of Korsakoff is a lack of insight into their disease. Therefore, some people with Korsakoff do not realize they cannot conduct certain tasks. Paul did not want to cause feelings of distress by explaining why such a *"childish and simple application was necessary"* (Paul) for Theo.

The use of STAPP was integrated into Theo's personal care plan, ensuring it contributed to Theo's day-to-day care, however, not everyone used the application. To be able to facilitate autonomy, there is a skillset required from the formal caregivers in using technology. One of the caregivers mentioned that *"it seemed fun, that people can do something by themselves without our help. Firstly, I thought it would be nice, but I was not looking forward to using it. I am not so good with computers"* (Cindy). She shows a sense of insecurity in the formal caregivers' skillset related to her digital literacy, as Cindy in general struggles with using technology.

### 5.2 Lacking and Missing In-person Support

The formal caregivers also expressed their concerns regarding the lack of in-person support. For example, Paul expressed that he does not feel comfortable letting Theo alone using STAPP: *"If I would ask a client to make coffee, I feel like I should stay nearby. And I will not do the laundry with someone else"* (Paul). This was agreed upon by Lucy: *"When something might go wrong, a person can interfere. So, like I said, when you are present in the kitchen, you can keep an eye on him"*. Interestingly, the observations revealed that when Paul and

Theo became more familiar with the application, Paul's guidance would become more passive as he started doing other care tasks nearby (e.g., making lunch) while Theo was making coffee.

In addition, trust, and skills in using technology in formal caregivers are needed to support them and relieve work pressure. For example, Lucy displayed a distrust in using STAPP and a fear that technology might take over enjoyable, social activities: *"I am not a fan of technology; I prefer human contact. That's why I haven't used it yet"* (Lucy). Paul also mentioned that, during their education, formal caregivers learn that personal guidance is very important and therefore would *"(. . .) not let a phone take over so easily"* (Paul).

Providing in-person support also involves formal caregivers reading and responding to specific responses and emotional moods of residents with Korsakoff. For instance, *"when someone makes a mistake, the resident must start over with the errorless learning method. The application does not do this, it cannot see whether someone performs a task (in)correctly. Someone must be returned to step one after a mistake. This cannot be substituted by a phone. Therefore, the human contact is missing"* (Paul). In addition, Paul mentions that *"(. . .) You still need the personal support, the humane aspect. Humans can see how someone feels. The application mentions a step in a monotone voice, but cannot see how someone feels, or when someone is tense"*.

### 5.3 Errorless Learning and the Importance of In-context Instructions

The application steps should be tailored to a specific household appliance. For example, the coffee machine in the care ward had a particular way of turning on (i.e., the 'on' button had to be pushed twice when the machine was turned off, but only once when the machine was on standby). This way of turning on the device was not incorporated in the application, and therefore, the formal caregiver had to intervene during task performance and Theo had to start over. These difficulties were illustrated with the laundry machine as well: *"Theo wanted to wash his bedding; however, this option is not given in the application and therefore he got stuck"* (Paul), and *"the laundry machine does not start when it is loaded incorrectly, the error is shown on the screen of the machine, but the application does not give a solution for this situation"* (C1). In addition, Paul mentions that doing laundry takes too long: *"The entire process is too long for residents to keep their focus. The app cannot solve this because personal guidance is necessary. The application says what someone must do but cannot see someone's emotions or whether someone is stressed"*, causing stress for Theo.

## 6 DISCUSSION

In this paper, we present a case study where we explored how a smartphone application (i.e., STAPP) can aid people with Korsakoff to execute daily tasks independently (i.e., make coffee, and do laundry). Simultaneously, we looked at how STAPP could support formal caregivers during their care tasks. From these findings, we learned three lessons: 1) assistive technology can help with increasing autonomy, 2) integration of technology in care practice, and 3) technology must be flexible and contain the ability to react in context and read emotions. This case study was insightful in retrieving these lessons. However, we believe that these lessons will also apply to a broader range of users such as individuals with

dementia or other (mild) cognitive impairments. Additionally, the contexts of these users will be different, for example, people in rehabilitation facilities or home care.

### 6.1 Increasing Autonomy and Individuality through Stepwise Guidance

The results suggest that mobile applications are a promising direction for increasing autonomy and help with re-learning ADL. Formal caregivers saw an added value to the concept of increasing autonomy and individuality for residents with Korsakoff because residents would be able to perform a task individually, without being corrected by a caregiver. In addition, Theo was able to make coffee without the help of Paul, enabling Paul to execute other tasks nearby and contributing to Theo's agency and maintenance of self [9]. Based on our findings, we identify an interesting direction for future research into how assistive technology that uses stepwise guidance can be introduced, and used, in a way that people with Korsakoff recognize the need for assistance and therefore may be motivated to use it. Additionally, we identify a direction for application development for stepwise-based activities in fields other than Korsakoff. This way, stepwise activities can be scaled, but this is not without limits as this restricts the ability to personalize the content.

### 6.2 Relying on the Digital Skills of Formal Caregivers

Interestingly, there is a contradiction between the aim for increased autonomy and the need for in-person support from formal caregivers. As we can see in the results, formal caregivers were enthusiastic about STAPP, and its potential for residents' autonomy. While testing, however, there were only a few staff members that used the application. We learned that involving caregivers is valuable for the adaptation of new technology in care practice [19] and that we need to consider the contexts in which the technology is used, and the role of key stakeholders [20]. Through the feedback of both Cindy and Lucy, we saw that the mobile application was difficult to integrate because of 1) insecurity in the formal caregivers' skillset related to digital literacy, 2) concerns regarding the lack of in-person support, and 3) a fear that technology might take over enjoyable, social activities. From Paul we learned that he 1) wanted to stay near Theo when he was using STAPP, and 2) that the formal caregivers are often the initiators of interacting with technology. These findings indicate that there is a threshold for trust and experience in technology [2] which is a reason why Paul stayed close while using the application [18]. This makes the usage of a mobile application dependent on the motivation, trust, and digital skills of the formal caregivers to use the product [20]. The findings indicate that there are different aspects in play (e.g., multiple stakeholders, different contexts, digital skills) that need to be considered while designing and developing assistive technologies, especially when the technology is introduced in different contexts.

### 6.3 Context-dependent and Flexible Technology

Even though modern technologies require more intensive training and cognitive demands on the user, casting doubt on the feasibility [13] and causing feelings of intimidation [18], we saw that Theo

was able to use the mobile application, illustrating that people with Korsakoff can use a slightly complex electronic aid such as a mobile phone [13]. The results, however, do reveal challenges for assistive technology for people with Korsakoff as 1) mobile applications should contain videos that require a specific amount of detail (i.e., specific choice of words, and clear communication and instructions), 2) are context-dependent, but 3) should also be feasible for other contexts. For instance, when the coffee machine would break down and be replaced with a different model, residents would not be able to operate the appliance anymore as the videos would not align. One solution is that caregivers can upload new activities with corresponding videos into the application [3]. This allows formal caregivers to tailor activities to the skills and learning needs of their clients with Korsakoff. For instance, to operate the television, play a DVD, set the time, or lay the table.

The caregivers mentioned a need for in-person support in the digital application. This in-person support is necessary to understand the physical, social, and emotional context of a situation (i.e., when a resident is stressed, or when the appliance gives an error), or guide residents back to step one when a mistake is made. These findings echo other research [7] that highlights the importance of human assistance in high-stress situations. Therefore, we suggest that future systems should be designed to keep humans in the loop to provide human assistance when necessary. Moreover, we identify incorporating emotional reading and in-situ support in technology as a promising future research direction for using technology to see how a mobile application can react more context-dependent [7].

## 7 CONCLUSION

In this case study, we present initial explorative research of STAPP: an application to support people with Korsakoff by completing multistep activities (i.e., making coffee, and doing laundry) and thereby contributing to autonomy and personhood. We share the design process and look at how STAPP could support both formal caregivers and people with Korsakoff. We focus on the qualitative experience of assistive technology and expand the literature about this technology in a specific context, which is still in its infancy. From this real-life case study, we learned that assistive technologies could help with increasing autonomy and must be flexible and contain the ability to be context-aware and read emotions. We also learned that formal caregivers are in favor of increasing autonomy but missed the in-person support while using the mobile application. In addition, it is important to keep in mind the skillset of digital literacy of formal caregivers, and their fear that technology might take over enjoyable activities, creating a threshold for using technology. The results motivate researchers to dive deeper into the context of assistive technology and people with Korsakoff, and what is necessary to ensure integration of technology in a care setting. We hope that we can encourage designers and researchers to continue to investigate how to best support underrepresented target groups, such as people with Korsakoff.

## ACKNOWLEDGMENTS

We want to thank the Dutch care organization Pleyade, their employees, and their residents for their collaboration in this project.

## REFERENCES

- [1] Nicolaas J.M. Arts, Serge J.W. Walvoort, and Roy P.C. Kessels. 2017. Korsakoff's syndrome: A critical review. *Neuropsychiatric Disease and Treatment* 13, 2875–2890.
- [2] Karabi C. Bezboruah, Darla Paulson, and Jason Smith. 2014. Management attitudes and technology adoption in long-term care facilities. *Journal of Health Organization and Management* 28, 3: 344–365.
- [3] Roeline Biemond, Erik Oudman, and Albert Postma. 2022. The Use of an Errorless Learning Application to Support Re-Learning of (Instrumental) Activities for People Living with Korsakoff Syndrome. *Journal of Clinical Medicine* 11, 23.
- [4] Elizabeth B-N Sanders and Pieter Jan Stappers. 2008. Co-creation and the new landscapes of design. *International Journal of CoCreation in Design and the Arts* 4, 1: 5–18.
- [5] Robert Bowman, Camille Nadal, Kellie Morrissey, Anja Thieme, and Gavin Doherty. 2023. Using Thematic Analysis in Healthcare HCI at CHI: A Scoping Review. *Conference on Human Factors in Computing Systems - Proceedings*, Association for Computing Machinery.
- [6] Jan Dewing. 2008. Process Consent and Research with Older Persons Living with Dementia. *Research Ethics* 4, 2: 59–64.
- [7] Emma Dixon, Rain Michaels, Xiang Xiao, *et al.* 2022. Mobile Phone Use by People with Mild to Moderate Dementia: Uncovering Challenges and Identifying Opportunities Mobile Phone Use by People with Mild to Moderate Dementia. *ASSETS 2022 - Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility*, Association for Computing Machinery, Inc.
- [8] Jenny Dudman, Julie Meyer, Cheryl Holman, and Wendy Moyle. 2018. Recognition of the complexity facing residential care homes: A practitioner inquiry. *Primary Health Care Research and Development* 19, 6: 584–590.
- [9] Geraldine Fitzpatrick, Alina Hultgren, Lone Malmberg, Dave Harley, and Wijnand Ijsselstein. 2015. Design for Agency, Adaptivity and Reciprocity: Reimagining AAL and Telecare Agendas. *Computer Supported Cooperative Work: CSCW: An International Journal* 2015: 305–338.
- [10] R. Galvin, G. Bråthen, A. Ivashynka, M. Hillbom, R. Tanasescu, and M. A. Leone. 2010. EFNS guidelines for diagnosis, therapy and prevention of Wernicke encephalopathy. *European Journal of Neurology* 17, 12: 1408–1418.
- [11] Wan He, Daniel Goodkind, and Paul Kowal. 2016. *An Aging World: 2015*. Washington, DC.
- [12] Tatiana Iakovleva, Elin M. Oftedal, and John Bessant. 2019. *Responsible Innovation in Digital Health: Empowering the Patient*. Edward Elgar Publishing.
- [13] E. A. De Joode, M. P.J. Van Bostel, P. Hartjes, F. R.J. Verhey, and C. M. Van Heugten. 2013. Use of an electronic cognitive aid by a person with Korsakoff syndrome. *Scandinavian Journal of Occupational Therapy* 20, 6: 446–453.
- [14] Erik Oudman, Tanja C W Nijboer, Albert Postma, Jan W Wijnia, and Stefan Van Der Stigchel. 2017. Neuropsychologische behandelingen voor patiënten met het syndroom van Korsakov: Review. *Tijdschrift voor Neuropsychologie* 12, 1–11. Retrieved September 13, 2023 from [https://www.tvnp.nl/inhoud/tijdschrift\\_artikel/NP-12-1-01/Neuropsychologische-behandelingen-voor-patiënten-met-het-syndroom-van-Korsakov-Review](https://www.tvnp.nl/inhoud/tijdschrift_artikel/NP-12-1-01/Neuropsychologische-behandelingen-voor-patiënten-met-het-syndroom-van-Korsakov-Review).
- [15] Erik Oudman, Tanja C.W. Nijboer, Albert Postma, *et al.* 2013. Acquisition of an instrumental activity of daily living in patients with Korsakoff's syndrome: A comparison of trial and error and errorless learning. *Neuropsychological Rehabilitation* 23, 6: 888–913.
- [16] Erik Oudman, Tanja C.W. Nijboer, Albert Postma, Jan W. Wijnia, and Stefan Van der Stigchel. 2015. Procedural Learning and Memory Rehabilitation in Korsakoff's Syndrome - a Review of the Literature. *Neuropsychology Review* 25, 134–148.
- [17] Yvon Ruitenburg, Rens Brankaert, Maarten Houben, Minha Lee, and Gert Pasman. 2023. Happje: Stimulating Collaborative Cooking for People with Dementia. *Conference on Human Factors in Computing Systems - Proceedings*, Association for Computing Machinery.
- [18] Yvon Ruitenburg, Gert Pasman, and Rens Brankaert. 2022. One Step at a Time: Evaluation of a Step-By-Step Recipe Tool Designed for People with Dementia. *Dementia Lab 2022: The Residue of Design. D-Lab 2022. Design For Inclusion*, vol. 3, Springer Cham, 77–92.
- [19] Corina Sas, Nigel Davies, Sarah Clinch, *et al.* 2020. Supporting Stimulation Needs in Dementia Care through Wall-Sized Displays. *Conference on Human Factors in Computing Systems - Proceedings*, Association for Computing Machinery.
- [20] Jenny Waycott, Ryan M. Kelly, Steven Baker, Barbara Barbosa Neves, Kong Saoane Thach, and Reeva Lederman. 2022. The Role of Staff in Facilitating Immersive Virtual Reality for Enrichment in Aged Care: An Ethic of Care Perspective. *Conference on Human Factors in Computing Systems - Proceedings*, Association for Computing Machinery.
- [21] . Takehiko Yamaguchi, Déborah Alexandra Foloppe, Paul Richard, Emmanuelle Richard, and Philippe Allain. 2012. A dual-modal virtual reality kitchen for (re)learning of everyday cooking activities in Alzheimer's disease. *Presence: Teleoperators and Virtual Environments* 21, 1: 43–57.
- [22] Ce Zhong, Doenja Oogjes, Ron Wakkary, and Amy Chen. 2019. Designing computational materiality: A preliminary study to explore the lived experience with transtexture lamp. *Conference on Human Factors in Computing Systems - Proceedings*, Association for Computing Machinery.